Typed Engineering Evidence for Changing Inspection Requirements: Reuse, Renewal and Refusal

Chuansu Peng[a], Haoran Dong[a], Ruoxuan Wu[a], Peipei Zeng[b]

[a]*School of Aeronautical Engineering, Civil Aviation University of China, China*
[b]*Engineering Training Center, Civil Aviation University of China, China*

Abstract

An engineering service must decide whether existing statistical evidence remains applicable when inspection requirements or producer configurations change. We extend a preserved anytime localization producer with a typed interface for image acquisition, capture-time spatial association and report delivery. Requirements compile to explicit failure events; interval containment gives a pointwise evidence-transfer witness, and event-specific dependencies delimit permissible producer changes. Exact binomial calibration and a finite simultaneous family support the stated population bounds. Two prospective synthetic studies execute 12,800 independent inputs and 409,600 paired policy/service outcomes, with actual image rasters and a predeclared wider-ROI intervention. A complete procedural semantic baseline receives the same knowledge and matches the contract's decisions; a cached direct-renewal baseline remains stronger in configured support. At risk allowance 0.15, semantic reuse supports 24/192 and 26/192 requests, versus 30/192 and 32/192 for direct renewal. Complete reporting remains uncertified. Deduplicated cold event preparation evaluates 2,490,368 records for anchor reuse versus 5,373,952 for direct renewal; all methods receive request caches. The contribution is an executable, auditable account of when engineering evidence can be reused, renewed or refused, with explicit negative results and simulation boundaries. It is not a representation-exclusive accuracy gain, general drift detector or physical safety certificate.

Keywords: Engineering knowledge representation, Inspection requirements, Evidence transfer, Risk calibration, Anytime localization

1. Introduction

An inspection service has to answer an engineering question beyond whether its latest pose estimate is accurate: does the available evidence support the requirement that the consumer currently asks it to satisfy? A camera can acquire an image with adequate sampling density while its target coverage is insufficient. A localization update can be useful for navigation yet arrive too late to geotag an earlier image. A report can meet its delivery requirement without satisfying the complete acquisition-and-association requirement. Reusing a statistical statement after these requirements change therefore requires an interpretation of the event being certified, its producer and the evidence lineage.

Engineering knowledge representations make requirements, domain quantities and change dependencies explicit [1, 2, 3]. The decision considered here is whether to retain a calibrated event statement, re-evaluate archived calibration outputs, or refuse a request. This is an engineering evidence-management problem whose numerical producer is an anytime airframe-relative localization service. The application is not described as a digital twin, and a metadata match is not treated as evidence that an unknown deployment distribution remains unchanged.

The retained localization study already includes a rigorous control: a fully validated procedural system can reproduce the contract system's numerical and authorization results. Fixed2 also remains competitive on ordinary routes. Those observations motivate an additional question: what capability does explicit task-event knowledge add when the consumer's requirements or the producer's configuration change? Improved localization error is neither necessary nor sufficient to answer that question.

We develop a typed interface that compiles requirements into failure events, emits a pointwise event-containment witness, and records the producer dependencies needed for that particular requested event. A calibrated anchor can cover a family of weaker requirements without rescoring calibration records. A producer edit can retain such evidence only when the requested event's implemented functional dependencies remain invariant. Otherwise the system renews the evidence from applicable records or refuses it. The interval algebra, binomial inversion and dependency reasoning are established tools; the contribution is their executable engineering interface, its explicit transfer conditions and its evaluation under requirement evolution.

Two prospective studies add image acquisition, capture-time spatial association and report delivery to the retained geometric producer. They use actual synthetic rasters and pixel processing, independently generated inputs, unchanged numerical predictors, and a predeclared wider-ROI intervention. A complete procedural semantic implementation and a cached direct-renewal baseline receive the same information and confidence budget. Their equality, superior support or lower runtime, when observed, remains part of the result. The study distinguishes requirement reuse, producer-change applicability and numerical scheduling; it does not claim an exclusive advantage of graph storage or a physical robot risk guarantee.

The paper first specifies the engineering quantities and retained executor, then presents the typed event interface and its conditions for evidence reuse. The prospective design and results evaluate service support, refusal, dependency propagation, actual record re-evaluation and computational trade-offs. Historical source and route results are retained separately to establish what the localization producer can and cannot provide.

2. Related work and scope of contribution

Engineering change analysis already formalizes relationships among requirements, behavior and structure. Wang et al. use Semantic Web representations and reasoning rules to propagate changes in a SysML system model [2]. Petnga and Austin incorporate temporal, spatial and dimensional semantics in knowledge-based cyber-physical decision support [1]. Ma and Hartmann explicitly connect inspection environments and defects to robot hardware requirements [3]. These works motivate explicit domain quantities and dependencies. Our endpoint is narrower: determining the applicability of a finite-sample task-event statement after a requirement or producer change. We do not claim a new general ontology, a human productivity experiment or semantic interoperability with external CAD standards.

Contracts for system design provide established refinement and composition principles [4]. Here, interval containment supplies a concrete refinement witness, while calibrated failure bounds are evidence attached to the frozen producer/population event. The proof does not derive stochastic reliability from a software type or component fingerprint. It requires both a valid statistical card and the stated event-transfer assumptions. Dependency equality concerns the specified program's data flow; fingerprints are not remote attestation of a physical device.

Distribution-free risk calibration can operate without refitting a predictive model. Learn then Test connects calibration to multiple testing [5]; conformal risk control treats broader losses [6]. Recent preprint work also studies non-monotone, multidimensional losses [7]. We retain exact binomial bounds for binary engineering events and a finite simultaneous family. Our containment argument reuses an already valid event bound; it does not replace those calibration methods or establish a new generic risk-control theorem. In particular, an after-observation choice among different calibrated anchors does not inherit a uniform risk bound merely by choosing the smallest upper bound.

Point-cloud registration and degeneracy-aware localization remain important numerical foundations [8, 9, 10, 11]. The retained solver and predictors are held fixed in the new studies. Numerical publication, source replacement and motion replay are separate from evidence authorization. This separation permits a fair test of task knowledge without attributing an unchanged estimator's performance to the representation layer. The strongest procedural comparator implements all current semantic and dependency rules; a result shared by both implementations is attributed to those rules rather than to their storage format.

3. Retained numerical producer and new inspection outputs

3.1. Immutable source, checkpoint policy and causal replay

The retained airframe-relative state is $x = (t, R, b, m_t, m_R)$ with an 18-dimensional local covariance. Measured motion increments propagate pose, bias and covariance. A scan and its source-time prior form one immutable source. Robust point-to-plane registration [9, 12] exposes checkpoints at up to 1, 2, 4, 8, 16 and 25 iterations. Each candidate uses the same scan/prior; later versions replace earlier versions rather than introduce another independent likelihood factor. At an already acquired receiver time t_d, a delivered candidate is assigned after replay of the measured increments:

$$(x_{t_d}^{(j)}, P_{t_d}^{(j)}) = \mathcal{R}_{s \to t_d}(x_s^{(j)}, P_s^{(j)}; u_{s+1:t_d}). \tag{1}$$

Ordered rollback installs the source prior after a rejected completion. Original source/prior seals, producer records, version ordering and history checks are retained. Their source-replacement and exposure proofs, detailed solver objective and original experiments remain in the supplementary and archival material.

The frozen CART and random-forest forecasts predict future endpoint error, event frequency and remaining work. Numerical selection trades predicted four-second error, failure cost and computation, with common stage-budget reservation. The primary new catalogue includes Fixed2, Fixed4, Final, FlatCART, FlatRF, KG1, ContractRF and OldRAVC. All receive the same current source, measured increments, numerical predictors, prices, budget wrapper and service load. KG1 adds the retained time/profile predicates; its name does not denote a learned graph embedding. The predictors are not refitted to the new camera tasks. The stronger procedural evidence baseline receives all task knowledge, rather than an artificially weakened numerical producer.

One actual common solver prefix is executed for each synthetic input. The separately executed policy/service wrappers consume only their revealed checkpoints and charge the corresponding prefix. This controlled trace design is paired counterfactual execution, not 32 independent inputs or 32 separately timed complete cold solves. Native component time and the deterministic work clock have different measurement boundaries. Historical full-route online timings remain a separate evidence block.

3.2. Actual raster and diagnostic target processing

A prescribed nominal route and prior CAD mesh plan up to 15 face/barycentric target addresses before stochastic map displacement and sensing. A synthetic gimbal aims at a prior-CAD patch. Targets remain the same across the paired ROI intervention. Raycasting against the perturbed geometric scene creates RGB pixels with procedural texture, fixed lighting, projected 25-mm-radius diagnostic landmarks, an exposure-blur surrogate and Gaussian intensity noise. A nearest-palette decoder estimates landmark pixel centers using only the image and known landmark colors. Ground truth is used for scene generation and offline correctness labels, not for decoding or online numerical policy choice. This is a diagnostic synthetic landmark task, not crack detection or a new real-camera experiment.

At a target with range d and absolute ray/normal incidence γ, the defined geometric sampling density is $g = d/(f_{\mathrm{px}}\gamma)$. Fixed-target quality uses the maximum GSD and motion/exposure blur across the preplanned addresses, including targets outside the current ROI. Correctly acquired target coverage is scored from actual decoded pixels. The capture-time geotag projects the CAD targets using the causally installed localization state. Its pixel discrepancy and metric landmark displacement are separately scored. An absent positive source version fails the readiness requirement; a later update cannot retroactively count as capture-time ready.

Image capture occurs at an independent, before-sensing sampled tick $c \in \{0.1, 0.2, 0.3\}$ s. Report delay and image age are distinct:

$$t_{\mathrm{report}} = \max\{t_{\text{first positive}}, c + 0.05 + \alpha w_{\mathrm{camera}}\}, \qquad a_{\mathrm{image}} = t_{\mathrm{report}} - c. \tag{2}$$

The camera clock characterizes the executed *virtual raster producer plus decoder*, not a physical camera's capture latency. Its excluded-pilot median is frozen separately for each ROI. The same declared load

Table 1: Engineering quantities and the decisions they support. All task thresholds are synthetic study specifications, not aircraft inspection or clearance standards.

Quantity	Unit	Interpretation and dependency
Fixed-target GSD	m/pixel	Sampling density at preplanned CAD addresses; focal length, target geometry and incidence
Exposure blur	pixel	Declared motion/exposure surrogate at the same targets
Target coverage	fraction	Correctly acquired diagnostic landmarks divided by planned targets; raster, ROI and decoder
Reprojection error	pixel	Causal estimated target projection versus decoded image landmark
Association error	m	Estimated CAD landmark displacement from its simulated physical counterpart
Localization readiness	Boolean	Positive, causally installed source version at image capture
Report delay; image age	s	Delivery time from source acquisition; delivery time minus image capture

multiplier is used across methods. Native raster generation, decoding and localization components are reported separately; their sum is a mixed-boundary virtual workload, not embedded online CPU or worst-case timing. This narrow timing model is part of the certified simulation event and cannot be transferred directly to physical inspection.

4. Typed engineering tasks and evidence transfer

4.1. Knowledge objects and trusted roles

We extend the retained structure $\mathcal{K} = (\mathcal{V}, \tau, \mathcal{R}, \sigma, \Gamma)$ with Quantity, Requirement, CompiledEvent, ProducerComponent and TransferWitness. Quantities carry a dimension and canonical unit; requirements reference quantities and admissible intervals. Relations include *measured-by*, *depends-on*, *compiled-to*, *covered-by* and *calibrated-on*. Immutable specifications identify the source, complete policy, population, event and calibration manifests. Executable predicates check units, event containment, component dependencies and card validity. The implementation uses typed records and a dependency specification rather than OWL or a learned graph embedding.

Observed pixels and measured motion are data. ROI, exposure, focal length, budget and the named population are declared configurations. Equality of deployment and calibration distributions is an assumption. The interface keeps these roles distinct: a successful configuration check cannot establish the distribution assumption.

The GSD value is stored in meters per pixel, with pixel treated as a discrete sampling index; its threshold conversion is between length units at that fixed sampling definition. Pixel-domain errors remain dimensionally distinct from metric errors. The compiler rejects incompatible units, unknown quantities, Boolean thresholds, nonfinite thresholds and empty intervals. Missing required outputs or nonfinite measured/scored values fail the event rather than satisfy a one-sided inequality.

4.2. Requirement compilation and pointwise coverage

Let $z_{p,j}(X)$ be quantity j produced under complete producer p on input X. A requirement R defines the admissible set

$$S_R = \{z : \ell_{R,j} \le z_j \le u_{R,j}, \quad j \in J_R\}, \qquad E_{p,R} = \{z_p(X) \notin S_R\}. \tag{3}$$

Absent lower or upper bounds are represented as unbounded intervals, while nonfinite outputs are explicitly outside S_R. An anchor A covers request R when $J_R \subseteq J_A$ and, for every requested quantity,

$$\ell_{A,j} \ge \ell_{R,j}, \qquad u_{A,j} \le u_{R,j}. \tag{4}$$

The compiler records each interval comparison as a witness. Consequently $S_A \subseteq S_R$ and $E_{p,R} \subseteq E_{p,A}$. Dropping an acquisition-quality requirement can therefore produce a covered *partial service*; it cannot be

advertised as satisfying the original complete inspection task. Adding a new required quantity or tightening an interval can require renewal.

The report event combines the required acquisition, association and timing violations using OR on the same input. Its calibration is a direct joint-event calibration. Two passing marginal cards are not presented as a passing joint card.

4.3. Event-specific dependencies under producer changes

Each quantity has a frozen implemented dependency set D_j. The requested event's closure is $D_R = \bigcup_{j \in J_R} D_j$. A component snapshot ν_p defines the capsule

$$v_{p,R} = H(\{(d, \nu_p(d)) : d \in D_R\}). \tag{5}$$

The capsule is a compact identity for a specified dependency projection. Its validity follows from the implemented data flow and trusted frozen code, not from hash equality alone. Transfer from producer p to p' requires that the requested quantities be identical functions of the same complete input: $z_{p',j}(X) = z_{p,j}(X)$ for $j \in J_R$. Matching dependency snapshots are the executable check for that narrowly specified condition.

A wider raster ROI changes coverage, pixel associations and the declared camera-work clock. It does not change geometric GSD at the same preplanned target addresses when pixel focal length and target geometry remain fixed. A GSD-only or fixed-target optical request can thus retain evidence through the verified projection; a complete image/report requirement generally cannot. Changing focal length or the target planner invalidates this optical projection. Deadline edits that affect capture, exposure, solver actions or processor scheduling change the producer and cannot be disguised as consumer-only relaxations.

The card binds the anchor event, old producer capsule, named population, n, k, allocated error budget, exact upper bound and complete calibration-manifest lineage. Invariant pooled cards identify both independent source blocks. Producer contexts are generated from the trusted frozen protocol; received metadata cannot rewrite the old context to manufacture a transfer witness. The retained source/prior sealing, causal version replacement and history refusal govern numerical attachment separately.

4.4. Admission, renewal and the fair procedural implementation

Before sensing, a request fixes its population, service, compiled event and risk allowance δ. A candidate policy is eligible if a valid applicable card covers the request and has $U \le \delta$. The selector minimizes the same calibration objective $\overline{L} + \lambda_{\text{sel}} \overline{W}$, with $\lambda_{\text{sel}} = 1\ \text{m}^2$, and policy-ID tie-breaking. No confirmation label or observed current-image quality chooses the policy or anchor. An empty set refuses the statistical task service; refused requests remain visible in service-support denominators.

`ContractCover` uses explicit containment witnesses and event capsules. `ProceduralCoverValidated` independently implements the complete interval comparisons and dependency checks with the same quantities, cards, caches and objective. `DirectRecertCached` evaluates each predeclared requested event on applicable cached calibration records; it receives invariant pools whenever the same dependency proof permits them. `ExactCard` is an exact-anchor matching control. The latter is a mechanism ablation rather than the primary competitive baseline. Cold preparation, first updates and repeated lookups have separate accounting, because a warmed direct-renewal table may be faster.

5. Conditions for evidence reuse

The source-replacement and delivery-exposure properties of the retained executor remain applicable under their original assumptions (Supplementary Material). The following propositions specify the additional task-event interface. They apply to a trusted frozen program and a stated population; they do not establish physical safety, detection of hidden drift or a new optimality guarantee for the learned numerical policy.

For a fixed binary anchor event, let k of n independent complete calibration inputs fail. With family error β and reserved family size K, store the exact one-sided bound [13]

$$U(k,n) = \begin{cases} 1, & k = n, \\ \text{Beta}^{-1}(1 - \beta/K; k+1, n-k), & k < n. \end{cases} \tag{6}$$

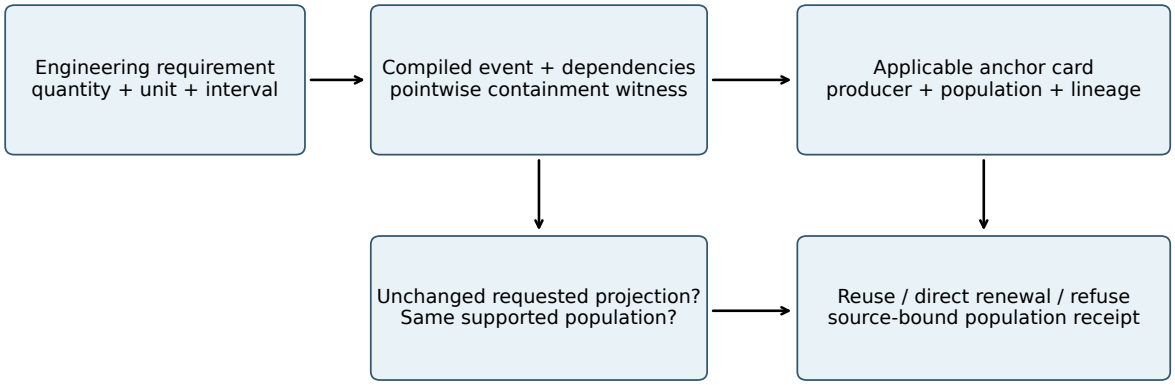


Figure 1: Executed engineering evidence interface. The dependency projection and event-containment witness provide separate checks. Matching metadata alone does not establish equality of calibration and deployment distributions.

The implementation evaluates the inverse survival function to avoid subtractive loss at a small allocated error probability. Validation checks integer counts, the exact numerical inversion, event identity and manifest lineage.

Proposition 1 (Uniform coverage of a requirement closure)**.** *Freeze the anchor producer/policy/population combinations independently of calibration outcomes. Assume iid complete calibration inputs and the same deployment distribution and event map. With probability at least* $1-\beta$ *over calibration, every stored anchor risk is at most its bound in Eq.* (6)*. On this confidence event, for every requirement* R *covered pointwise by any such anchor* A, $\Pr(E_{p,R}) \le U_{p,A}$*. This conclusion is simultaneous over the entire containment closure, including requirements not individually evaluated on calibration records.*

Proof. For each fixed anchor the binomial inversion has undercoverage at most β/K. A union bound over the reserved finite card family establishes simultaneous anchor coverage, without requiring independence across policies or events. Equation (4) gives $E_{p,R} \subseteq E_{p,A}$ pathwise. Taking probabilities on the simultaneous confidence event establishes every covered requirement bound. No union bound over those derived requirements is needed. □

Policy selection using calibration objectives and valid simultaneous bounds is permitted. The selected complete policy remains fixed across the declared population for that before-sensing request. The confidence event does not make the bound conditional on observed image quality or on receiving a particular pose.

Proposition 2 (Transfer through an invariant requested projection)**.** *Let an old producer* p *have a valid anchor bound* $U_{p,A}$*. Suppose* A *covers* R*, the complete-input population is unchanged, and the implemented requested quantities satisfy* $z_{p',j}(X) = z_{p,j}(X)$ *for all* $j \in J_R$ *and all supported complete inputs. Then* $\Pr(E_{p',R}) \le U_{p,A}$ *on the same confidence event, even if producer* p' *changes quantities not required by* R*.*

Proof. Requested-projection equality implies $E_{p',R} = E_{p,R}$, including the stated nonfinite-output semantics. Pointwise coverage implies $E_{p,R} \subseteq E_{p,A}$. Combining these two relations with the old anchor bound proves the result. □

The premise is a functional invariance condition, stronger than merely matching metadata. Our implementation checks a frozen, inspected dependency projection and tests paired outputs for the predeclared ROI intervention. It does not infer general distribution invariance from finitely many passing samples. Changing an omitted component is permissible only within the implemented functional model; an undeclared physical coupling invalidates that interpretation. Pooling independent calibration draws from two such invariant producers is valid for that event only, and both raw-manifest identities are retained.

5.1. Why adaptive anchor switching needs another guarantee

Two events can each have risk 0.10 while their union has risk 0.20. For example, let disjoint sets of inputs trigger the two events, and choose the corresponding unfavorable event after observing the input. A marginal certificate for each event does not bound the selected event by 0.10. The same issue affects per-image policy or anchor switching. A common covering event, a directly calibrated switched policy, or an applicable union bound is needed. The studied selector commits before opening the source and does not make that switch.

5.2. Applicability and refusal

A request without a valid covering card or successful renewal is refused. A malformed source or a detected invalid version attempt invokes the retained source-history refusal mechanism. A request that changes capture, execution budget or scheduling cannot reuse an old card solely because a textual deadline is relaxed. Unknown operating distributions remain unsupported. Nor does an episode statement imply a route statement: without additional structure, M episodes with individual allowance δ yield at most the generic union bound $\min(1, M\delta)$. These conditions delimit the engineering service rather than hide its unsupported regions.

6. Prospective design and reproducibility

6.1. Two studies, freezes and independent units

Study A uses a 192×144 raster; the predeclared Study B intervention uses 288×216 at the same 1,800-pixel focal length and 6-ms exposure. Wider ROI therefore changes the set of acquired pixels and virtual production cost while preserving geometric GSD at the same target addresses. B uses independent input seeds and its changed non-invariant events require fresh calibration. Invariant optical events may use the explicitly identified independent A/B calibration pool, available equally to every baseline.

Each study contains 2,048 nominal and 2,048 unfitted-geometry calibration inputs, 1,024 independent confirmation inputs for each population, and 64 inputs in each of four unsupported stress profiles. Thus each study has 6,400 independent inputs and the two studies have 12,800. The 32 policy/service outcomes per input do not increase sample size. Forty-eight unique pilot seeds are excluded; iterative pilot implementations repeat these same seeds. Additional excluded execution fixtures reuse named seeds and do not become calibration observations.

The nominal mixture samples B767/C172 identities, source frames, prior deviations $\{0.015, 0.03, 0.06\}$ m, range-noise multipliers $\{1, 2\}$ and shape levels $\{0, 1, 2\}$. The unfitted procedural airframe is excluded from forecast fitting and receives its own matched evidence. Map translation and rotation perturbations have standard deviations 0.025 m and 0.001 rad. Stress separately removes a 2.2-rad range sector, retains a 0.5-rad sector, increases prior deviation to 0.08/0.12 m, or applies a 0.06-m map warp. Stress profiles have no primary cards and cannot inherit matched-population certification.

Numerical algorithms, camera/task modules, thresholds, seed domains, dependencies and timing were frozen before formal input generation. A disclosed format amendment, still with zero formal inputs, normalized manifest paths and preserved hashed JSON bytes across Git/Windows checkout; it changed no scientific parameter. The archived initial freeze and amendment are retained. All calibration cards and choices for both studies are then sealed before any confirmation generation. These are locally archived prospective records with public GitHub checkpoints, not an independently registered trial.

Windows CPU observations show 15.625-ms quantization. Medians characterize the declared virtual-service clock; distributions of native costs remain reported rather than interpreted as precise embedded component latencies. The historical localization operation clock is preserved independently.

6.2. Requests and fair competitors

The full acquisition task requires GSD at most 20 mm/pixel, blur at most 0.5 pixel and correctly acquired target coverage at least 0.50. Association requires coverage at least 0.50, reprojection at most 30 pixels, metric displacement at most 0.10 m and a positive source version at capture. Full reporting requires the

Table 2: Shared numerical services and new virtual camera clock. Camera cost is additional to the retained numerical budget and defines Eq. (2); it is not physical camera latency.

Service	Load α	Numerical deadline (s)	Stage budget (ms)
Relaxed	10	0.5	20
Standard	20	0.3	12
Tight	40	0.2	6
Contended	60	0.3	12
A: raster 46.875 ms; decoder 15.625 ms; total 62.5 ms			
B: raster 78.125 ms; decoder 46.875 ms; total 125 ms			

union of these constraints, report delay at most 2 s and image age at most 1.8 s. These are synthetic diagnostic-service specifications.

Eight predeclared variants include coarser GSD, larger tolerated blur, explicitly partial coverage, a further relaxed partial request, stricter requirements, later reporting and a single-component service. Full, partial and single-component events form the anchor set. The entire matrix is retained; removing requirements produces a different service, not success on the full task. The primary allowance is 0.15, with 0.10/0.20/0.25 secondary settings.

The reserved simultaneous budget is $\beta = 0.05$, $K = 4,608$, covering both studies, populations, numerical policies, services and the predeclared event families. Unused reserve is not reassigned after results are seen. Every method uses the same per-card allocation, applicable records, invariant pool, calibration objective and request cache. Direct renewal evaluates the actual requested binary event; it may support more requests than an available anchor closure. Procedural coverage implements the same complete semantics as the contract. Exact-card matching is an explicitly limited control.

6.3. Evaluation and preservation

We separately report configured service support, refusal, conditional confirmation failure, incorrect applicability, certificate margin, cold record evaluation, preparation and repeated-query cost. Refused requests contribute zero supported service and remain in every configuration-count denominator; no successful task-event sample is invented for an unexecuted refused policy. Numerical error and work are reported for all eight producers and separately for authorized selections. Same-source pairing and population-stratified summaries prevent repeated queries from becoming independent observations.

Native source arrays, measured increments, every executed solver checkpoint, RGB/depth rasters, decoded landmarks and publication/installation traces are retained with hashes. Continuation verifies raw input archives, not only result tables. Complete confirmation repetition spans four services; its row count measures executed queries, not evidence sample size. Historical A19/A20/R8 source, 36 R8 route inputs/432 executions, four original photographs and five supplied simulation views remain intact. No new physical trial or route-level task certificate is asserted.

7. Results and engineering interpretation

7.1. Configured service support and complete-task limits

The complete catalogue contains 192 task/variant/population/service requests per study and allowance. Figure 2 and Table 3 retain unsupported requests in the denominator. At the primary allowance, the two semantic implementations support 24 requests in A and 26 in B. Direct renewal supports 30 and 32; exact-anchor matching supports 18 and 20. The equality of the contract and complete procedural implementation holds for all 6,144 frozen plans and their repeated confirmation outputs. The semantic layer therefore supplies a transferable interpretation of evidence, but the results do not establish an advantage exclusive to its representation.

At allowance 0.15, no complete acquisition task is certified in either study. The only full-task semantic plan is B/unfitted/relaxed association, using FlatCART with upper bound 0.143973. Its margin is 0.006027,

Table 3: Primary-allowance configured support. Each denominator retains every task, variant, population and service. Full-task support counts only the original complete requirement.

Study	Method	All requests	Full tasks
A	ContractCover	24/192	0/24
A	ProceduralCover	24/192	0/24
A	DirectRenewal	30/192	0/24
A	ExactCard	18/192	0/24
B	ContractCover	26/192	1/24
B	ProceduralCover	26/192	1/24
B	DirectRenewal	32/192	1/24
B	ExactCard	20/192	1/24

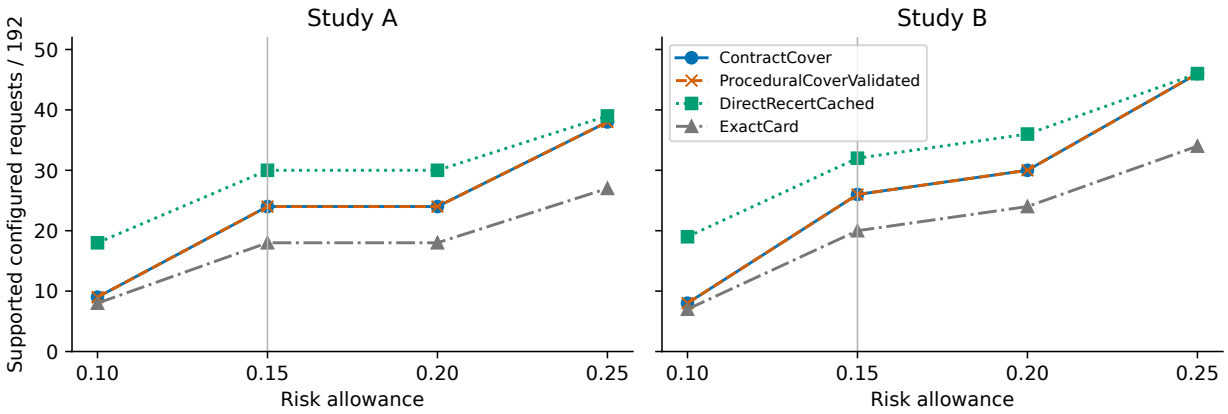


Figure 2: Configured support under the full request matrix, with a common confidence allocation, policy objective and records. The primary allowance is 0.15. Contract and complete procedural curves coincide. Direct renewal retains its additional supported services; exact-card matching is a limited mechanism control. These are configuration counts, not a field-request probability.

so its acceptance is narrow. Complete reporting remains refused in every population/service cell. Most supported requests are partial or single-component services, and several nominal variants compile to the same event. They must not be described as independent successful tasks or as recovery of the full inspection service.

The wider ROI adds two primary supported configurations, yet its larger virtual production cost limits timely reporting. Secondary allowances expose additional configured support, including B's increase from 26 to 30 at allowance 0.20; changing the allowed risk is not an improvement under the original 0.15 requirement. Figure 3 displays every primary request, including the direct-renewal-only services and the unsupported tight/contended region. A stronger direct baseline is retained instead of being constrained to the anchor family.

7.2. Optical intervention and causal association

Actual rasters are produced by raycasting and decoded without truth coordinates. Figure 4 shows the first preselected excluded pilot under the two ROIs. The 48 paired pilot sources verify identical fixed-target GSD: enlarging the ROI does not change range, focal length or incidence at the planned addresses. It can acquire more targets while leaving the sampling-density requirement unsatisfied. Figure 5 reports the independent confirmation distributions, separated by their calibrated populations. Study A and B confirmation samples use different seeds; their empirical differences are not a paired intervention estimate.

Association uses the numerical version available at capture. A later update cannot repair readiness retrospectively. Report delay includes the raster/decoder virtual-service clock; image age subtracts the independently sampled capture time. This distinguishes acquisition, association and delivery instead of

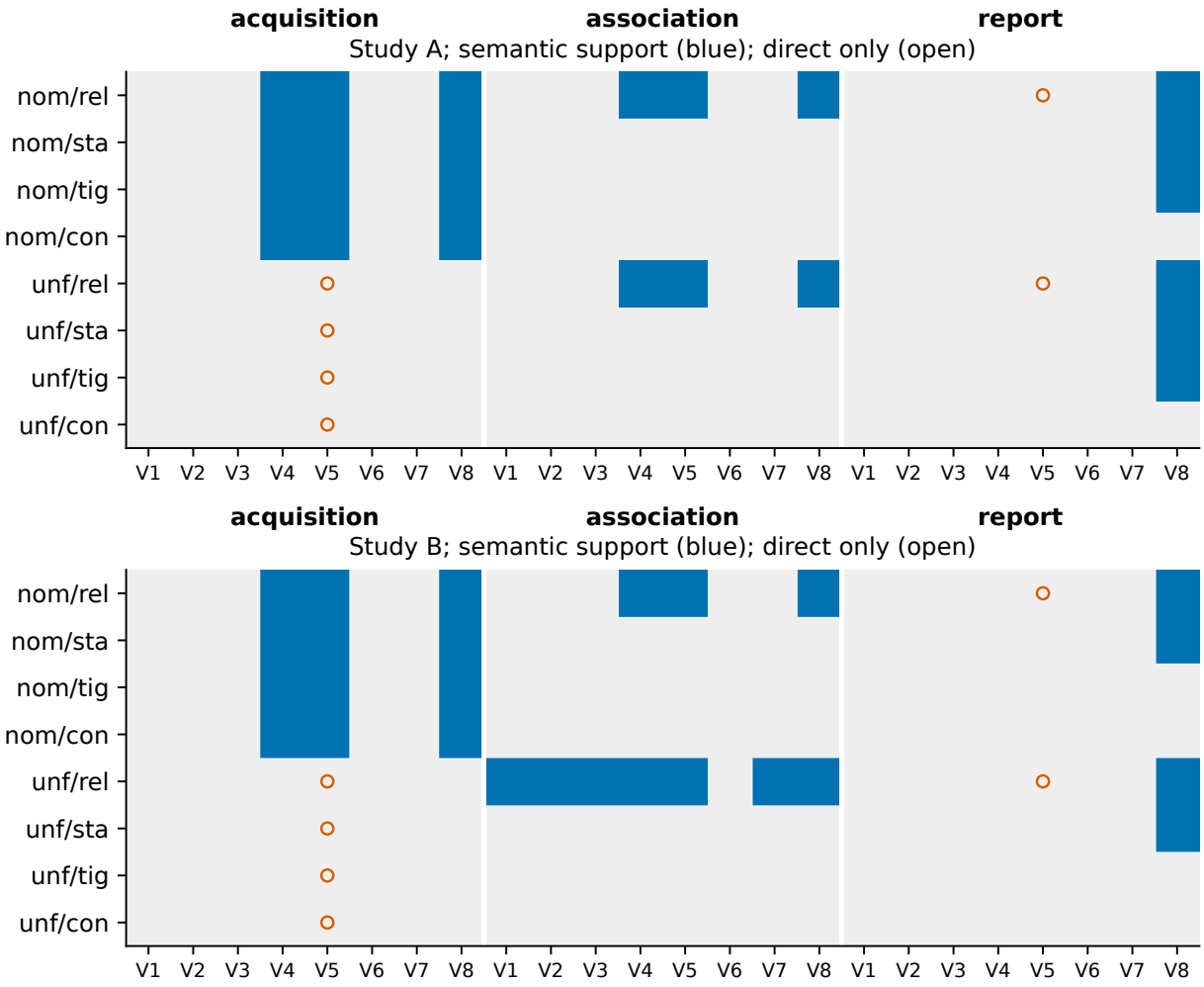


Figure 3: All primary configurations. Each study has eight population/service rows and 24 task/variant columns. V1–V8 denote full, coarser GSD, more blur, partial, partial relaxed, stricter, late report and single component. Blue indicates semantic support; open orange circles retain additional direct-renewal support. Compiled-event duplicates remain visible but do not increase evidence sample size.

measuring three copies of localization accuracy. The calibration and confirmation tables retain all task quantities and every policy/service outcome, including nonfinite-score failures.

7.3. Certificate stability and transfer boundaries

All 3,072 stored cards have been recomputed from their executed failure counts with the reserved 4,608-family allocation. Cards identify the population, requested or anchor event, complete producer dependencies and raw calibration lineage. Invariant pooled cards use 4,096 independent inputs; other cards use 2,048. The unused reserve is not redistributed.

Figure 6 retains a passing full-association anchor, a component optical anchor and an unsupported full optical event. Smaller nested prefixes frequently lose certification, and even a full-sample passing association card has limited primary margin. Prefix sizes and alternative confidence levels are diagnostic grids; they do not receive an additional simultaneous guarantee, change F1, or produce renewed confirmation-based selection.

Table 4: Fixed before-sensing plans and independent confirmation counts; the rows are representative configured services, not independent copies of the source sample. All plans are supplied as CSV.

Study	Pop.	Service	Task	Variant	Policy	U	Confirm. k/n
A	nom	rel	acqui	partial	OldRAVC	0.1472	110/1024
A	nom	rel	acqui	single service	OldRAVC	0.1375	98/1024
A	nom	rel	assoc	partial	FlatCART	0.1494	120/1024
A	nom	rel	assoc	single service	FlatCART	0.1128	82/1024
A	nom	rel	repor	single service	OldRAVC	0.0069	1/1024
A	unf	rel	assoc	partial	FlatCART	0.0733	54/1024
A	unf	rel	assoc	single service	FlatCART	0.0568	39/1024
A	unf	rel	repor	single service	FlatCART	0.0056	1/1024
B	nom	rel	acqui	partial	OldRAVC	0.1391	106/1024
B	nom	rel	acqui	single service	OldRAVC	0.1285	106/1024
B	nom	rel	assoc	partial	FlatCART	0.1095	90/1024
B	nom	rel	assoc	single service	OldRAVC	0.1423	123/1024
B	nom	rel	repor	single service	OldRAVC	0.0080	1/1024
B	unf	rel	assoc	full	FlatCART	0.1440	103/1024
B	unf	rel	assoc	partial	FlatCART	0.0750	43/1024
B	unf	rel	assoc	single service	FlatCART	0.0616	29/1024
B	unf	rel	repor	single service	FlatCART	0.0069	0/1024

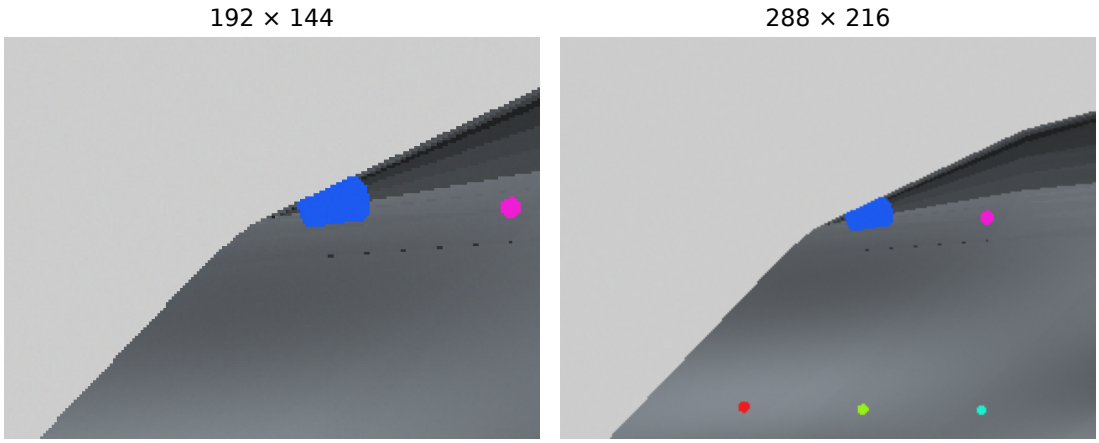


Figure 4: Actual rendered pixels for excluded pilot seed 12000000 under the predeclared ROI intervention. Targets and pixel focal length are fixed. These synthetic diagnostic landmarks illustrate the producer, not aircraft defects or real-camera observations. The same pilot source is used in both panels.

Evidence reuse through the ROI change is permitted only for the implemented invariant requested projection. It does not certify new coverage, association or delivery functions by attaching an unchanged optical capsule. Seventy-two excluded provider-change fixtures verify conservative invalidation after capture, motion or geometry provider edits. Thirty-two malformed receipt fixtures produce no incorrect authorization, maintain refusal across reconstructed gates, and admit independent trusted successors. These deterministic software fixtures demonstrate the stated applicability rules, not a probabilistic fault rate or detection of unknown physical drift.

The source-bound receipt observer runs against frozen plans and archived execution records. It binds the trusted source identity, source seal, population, study, policy, compiled event, plan and card. This is executed attachment bookkeeping after the numerical experiment, rather than an added real-time camera pipeline. Its measured cost is reported separately. The source/prior/version enforcement inside the retained

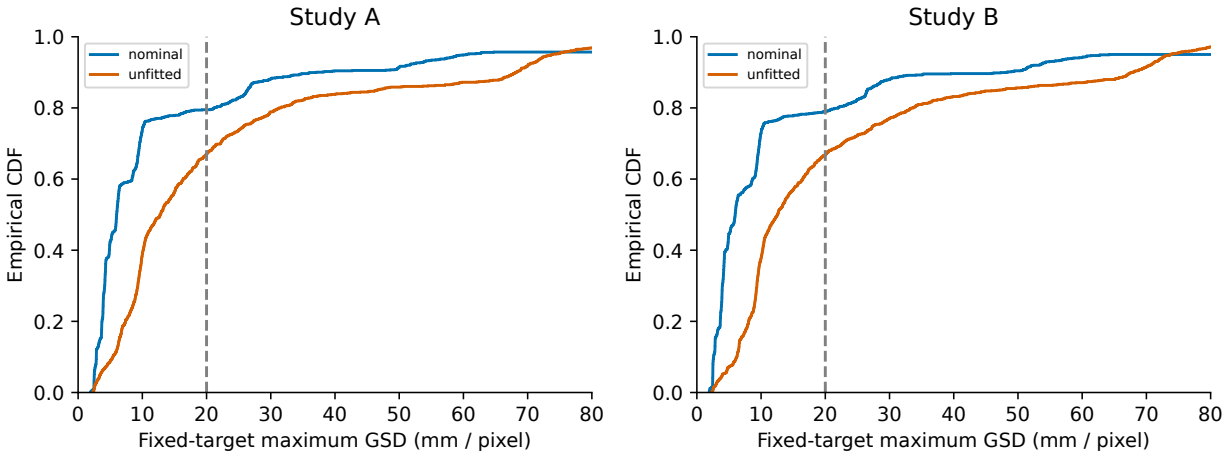


Figure 5: Independent confirmation empirical GSD distributions for the nominal and unfitted populations, with a 20-mm/pixel requirement reference. Fixed-target GSD is a geometric score and includes planned targets outside the acquired ROI. The functional ROI invariance is checked on paired pilots; the formal A/B samples are independent.

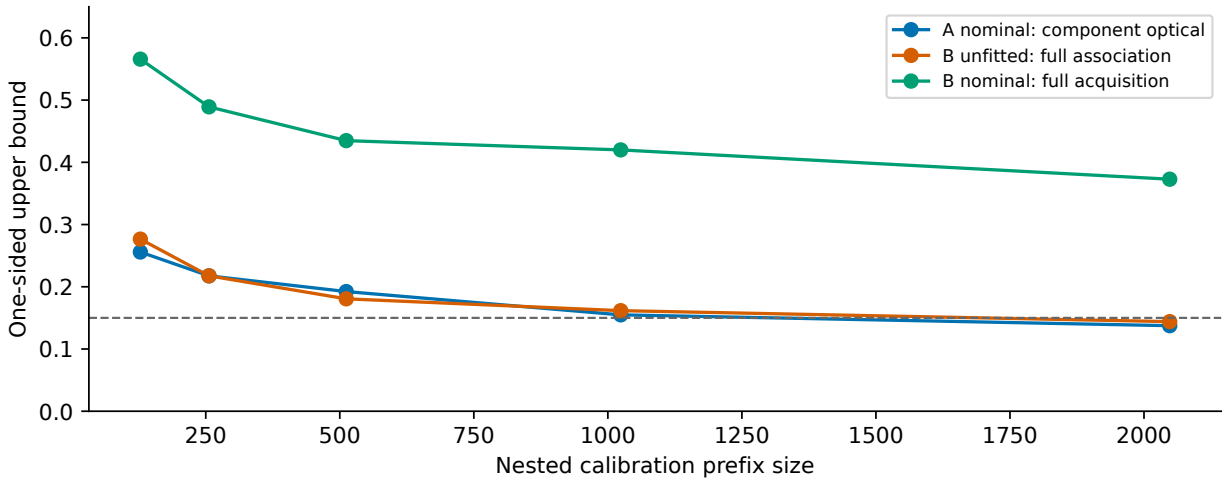


Figure 6: Actual fixed-event failure counts under nested calibration prefixes and the original reserved-family allocation. The 0.15 line is a requirement, not a fitted target. Displayed prefixes do not create independent samples or an extra guarantee over this sensitivity grid. Full-precision counts and confidence settings are supplied as CSV.

localization executor remains a different, actually executed component.

7.4. Executed update and cache costs

With event-identity deduplication available to every method, the anchor methods prepare 1,152 unique cards and the direct method prepares 2,560. Their record visits are 2,490,368 and 5,373,952 respectively, a 53.66% reduction for anchor reuse. This is a trade-off: less event re-evaluation supports fewer configured requests. The complete procedural method obtains the same reduction. Preparation wall time is measured on equally preloaded records; disk reads, numerical producer execution and human development are outside this comparison.

The first 384-request batch separately includes numeric-card validation and plan search. Every method then receives the same request dictionary cache. Warm lookups require no record rescoring, and their small timing differences are not treated as a scientifically exclusive speed advantage. Known direct-event duplicates

Table 5: Executed computation diagnostics with equally preloaded records and event-identity deduplication. Cold event preparation is the median of three trials; the first batch contains 384 primary queries. Warm time is per shared-cache lookup. These are not developer-effort or embedded worst-case measurements.

Method	Cards	Record visits	Prep. (s)	First batch (s)	Warm (μs)
ContractCover	1,152	2,490,368	0.179	6.408	0.091
ProceduralCover	1,152	2,490,368	0.176	6.225	0.111
DirectRenewal	2,560	5,373,952	0.365	9.991	0.098
ExactCard	1,152	2,490,368	0.186	8.747	0.103

are cached as well. JSON arrays are restored to the declared immutable context tuple fields before procedural re-query, so serialization does not artificially reject that baseline. All re-queried plans reproduce F1.

7.5. Numerical trade-offs, geometry and retained routes

Figure 7 retains all eight numerical policies, separated by B767, C172 and the unfitted model. The Pareto boundary is descriptive. Fixed2 is kept wherever its work or error trade-off is favorable; a representation claim does not remove a competitive fixed strategy. Complete numerical summaries include timely positive-version success, mean error, mean episode-tail error, 95th percentile of episode maxima, work, publications and native components. Population calibration remains on the frozen mixture, so the model-separated descriptive tables do not imply separately calibrated conditional certificates.

Unsupported occlusion, degenerate range sectors, larger priors and map warps receive no transferred population card. All 512 stress inputs are retained as descriptive evidence. The preserved R8 study additionally contains 36 complete-route inputs and 432 executions. Ordinary routes can favor Fixed2; C172 contention can improve KG1 timely success while worsening its RMSE. These archived route observations are not rerun or renamed as new R9 evidence. They do not certify a route-level image/report event. The complete adverse route tables, four real platform photographs and five original simulation views remain in the accompanying R8 archive and supplementary material.

8. Contribution, engineering value and limits

The new evidence concerns three kinds of engineering decisions. A consumer-only requirement relaxation can inherit an anchor bound when the complete failure event is pointwise covered. A producer edit can retain that bound only for a proven invariant requested projection. A remaining request requires direct renewal or refusal. The domain quantities, units, source lineage and dependency witnesses make these decisions inspectable. This is stronger than checking whether a stored card's label matches a request, but it is also implementable by a complete procedural system.

The strongest procedural comparator demonstrates that fact. It is given all dependency knowledge, event comparisons, records and caches and reproduces the contract's authorization and numerical outputs. Accordingly, the independent value shown here belongs to *explicit executable semantics*, rather than a graph data structure. The interface centralizes the interpretation and produces a checkable witness; no experiment establishes that engineers maintain it faster, that a graph database is required, or that generic programming cannot achieve it. Change reasoning and interval refinement already exist in engineering informatics. Our claim is their concrete integration with finite-sample event evidence and the measured inspection trade-offs, not invention of those general tools.

Direct renewal is also a serious competitor. It requires more event-record evaluations but supports additional requests and can cache subsequent queries. Semantic closure is useful when applicable anchors already exist and repeated relaxations matter; direct renewal is useful when sufficient applicable raw outputs exist or the anchor bound is conservative. An engineering implementation should permit both. Restricting the direct baseline's old-card access would weaken that comparison; the pre-confirmation parity amendment removes that restriction and preserves its prior supported services.

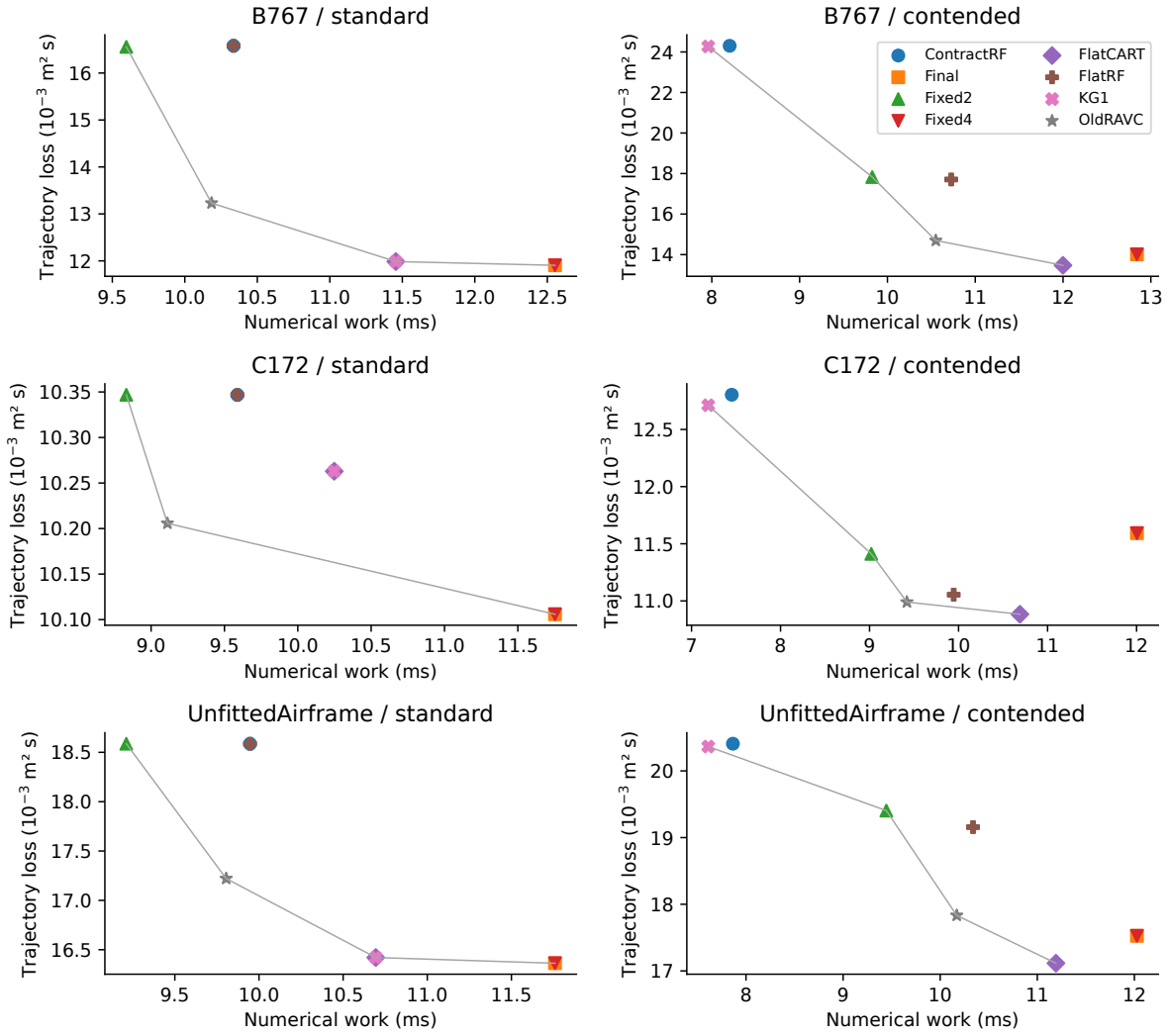


Figure 7: Study B confirmation numerical work–loss means, separately for each geometry at standard and contended services. All eight policies, including Fixed2, remain visible. Camera work is common across policies in each panel and omitted from the horizontal axis only to expose numerical scheduling trade-offs. Full total-work values are supplied. Nondominated means are not confidence bounds.

The ROI experiment demonstrates selective transfer rather than complete task recovery. A geometric optical quantity can remain invariant while target coverage and delivery cost change. The full reporting service remains uncertified at the primary allowance. Relaxing risk or deleting quality requirements defines a different service; the paper reports such alternatives explicitly instead of treating them as improvements to the original request. Fixed2's retained numerical trade-offs likewise prevent an unsupported claim of universal system-level superiority.

External validity remains limited. Synthetic palette landmarks are diagnostic targets, not natural surface defects. The virtual producer uses prescribed trajectories, declared perturbations and a frozen camera-work clock. Native CPU observations are quantized and depend on a desktop environment; they do not characterize an embedded camera's real-time operation. Real deployments can contain unmodeled lighting, texture, exposure-control, scheduling and mechanical couplings that invalidate a dependency projection. A metadata match cannot reveal such changes. New matched calibration, physical evidence and a suitable distribution-monitoring mechanism would be required before claiming physical reliability.

The three task families share one geometric source and executor. They establish a broader executable consumer interface than localization alone, but do not establish generalization to unrelated engineering domains. Deterministic malformed-source fixtures test specified refusal logic, not adversarial robustness in an open system. Complete-route performance remains historical descriptive evidence without mission-level task certification. These limitations are substantive: the study improves the evidence interpretation and cross-task validation, while the representation's broad originality and application reach remain questions for peer review.

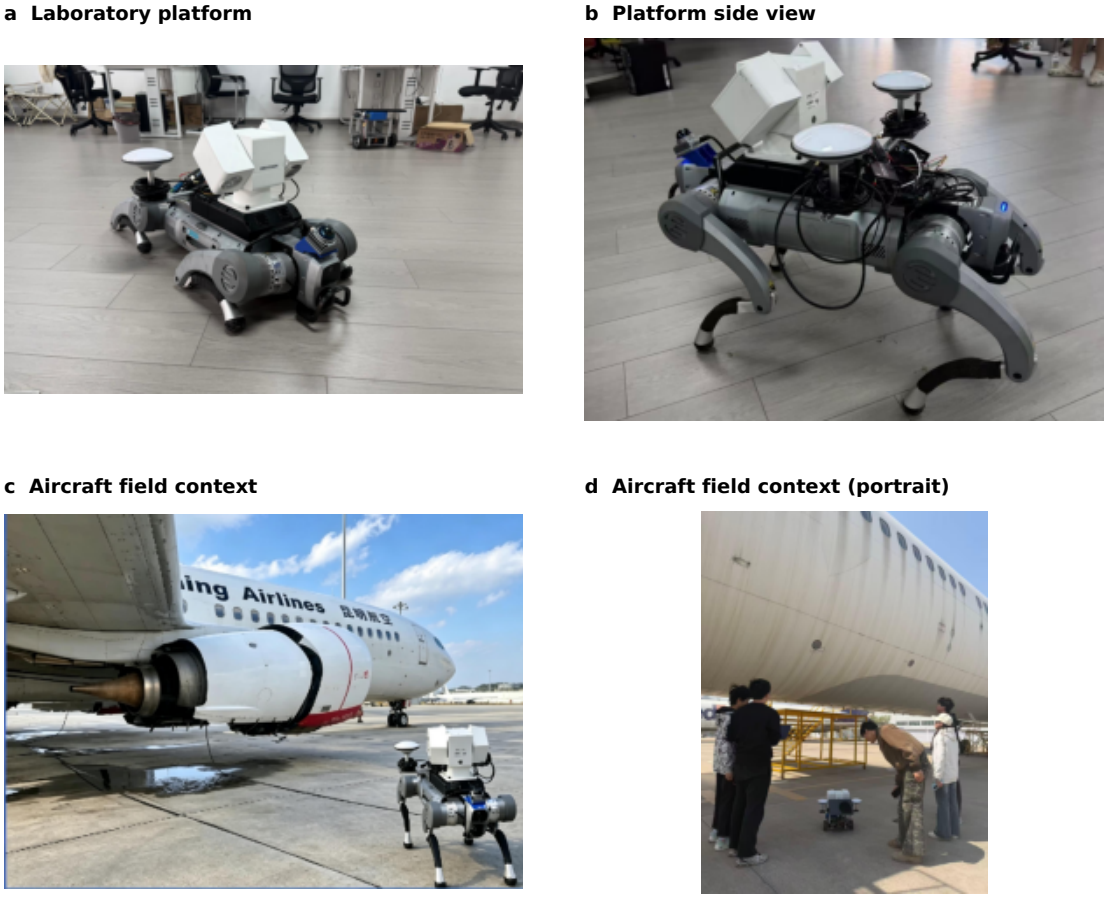


Figure 8: Four original real platform photographs preserved from the supplied research assets. They document engineering context and do not validate the newly simulated image, association or report events. Full original-resolution photographs and five prior simulation views remain in the historical release.

9. Conclusion

An executable task-event interface can delimit reuse of engineering evidence through a requirement containment witness and a trusted invariant dependency projection. Otherwise the service renews applicable evidence or refuses the requested claim. Two independently generated synthetic studies evaluate acquisition, capture-time association and timely reporting while retaining the original numerical solver, competitive fixed policies and adverse route results.

The complete procedural semantic implementation matches the contract, and direct renewal supports more configured requests. Anchor reuse reduces measured cold event re-evaluation but does not establish a

graph-exclusive accuracy or runtime advantage. The wider ROI changes acquisition and cost while preserving only a narrow optical projection; full reporting remains unsupported at the primary risk allowance. The release preserves all raw inputs, actual rasters, frozen choices, negative outcomes and code so subsequent engineering validation can extend these stated boundaries.

Evidence and applicability boundary

The new studies contain 12,800 distinct synthetic inputs: 8,192 calibration, 4,096 confirmation and 512 unsupported stress. Forty-eight pilot seeds, repeated development fixtures and five source replays add no independent evidence sample size. The 409,600 policy/service outcomes and 6,291,456 confirmation request rows are paired repetitions. Historical R8 inputs and 36 routes/432 executions remain a separate evidence block. Certificates concern unconditional frozen-policy/population events under iid sampling, matched distributions and trusted implemented event maps. They do not provide per-image conditional risk, unknown-drift detection, physical safety or embedded worst-case timing. The raster/decoder clock describes a virtual producer; receipt-observer and numerical timing boundaries are reported separately. No new real-robot experiment or route-level inspection certificate is asserted.

Funding

This work was supported by the Civil Aviation Safety Capacity Building Fund Project (RJ2025073), "Pilot-scale development of civil aircraft walk-around inspection and on-wing aero-engine cleaning equipment."

Data and code availability

Code, editable sources, result tables and handoff instructions are available in the public R9 research branch. Raw arrays, actual rasters, publication journals, phase manifests and SHA256-verified archives are available in its checkpoint release; the final delivery index identifies the complete reading package. The historical R8 release preserves A19/A20 sources, original data, route results, four real photographs and five original simulation views. Small reading and analysis packages permit continuation without downloading the entire raw dataset.

Declaration of generative AI and AI-assisted technologies

OpenAI ChatGPT/Codex (GPT-6) assisted research-code development and reconstruction, computational analysis, manuscript drafting and revision, literature organization, and code for exact schematics and data visualizations. Numerical observations were produced by the archived geometric simulation and executed localization code. The authors must review the work and confirm responsibility before submission.

References

[1] L. Petnga, M. Austin, An ontological framework for knowledge modeling and decision support in cyber-physical systems, Advanced Engineering Informatics 30 (1) (2016) 77–94. doi:10.1016/j.aei.2015.12.003.

[2] H. Wang, V. Thomson, C. Tang, Change propagation analysis for system modeling using semantic web technology, Advanced Engineering Informatics 35 (2018) 17–29. doi:10.1016/j.aei.2017.11.004.

[3] L. Ma, T. Hartmann, A proposed ontology to support the hardware design of building inspection robot systems, Advanced Engineering Informatics 55 (2023) 101851. doi:10.1016/j.aei.2022.101851.

[4] A. Benveniste, B. Caillaud, D. Nickovic, R. Passerone, J.-B. Raclet, P. Reinkemeier, A. Sangiovanni-Vincentelli, W. Damm, T. A. Henzinger, K. G. Larsen, Contracts for system design, Foundations and Trends in Electronic Design Automation 12 (2–3) (2018) 124–400. doi:10.1561/1000000053.
URL https://www.nowpublishers.com/article/Details/EDA-053

[5] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, L. Lei, Learn then test: Calibrating predictive algorithms to achieve risk control, The Annals of Applied Statistics 19 (2) (2025) 1641–1662. doi:10.1214/24-AOAS1998.
URL https://projecteuclid.org/journals/annals-of-applied-statistics/volume-19/issue-2/Learn-then-test--Calibrating-predictive-algorithms-to-achieve-risk/10.1214/24-AOAS1998.full

[6] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, T. Schuster, Conformal risk control, in: The Twelfth International Conference on Learning Representations, 2024.
URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/f3549ef9b5ff520a7e41ff3cc306ab2b-Abstract-Conference.html

[7] A. N. Angelopoulos, Conformal risk control for non-monotonic losses, arXiv preprint, version 1; not represented as a journal publication (2026). arXiv:2602.20151, doi:10.48550/arXiv.2602.20151.

[8] P. Besl, N. D. McKay, A method for registration of 3-D shapes, IEEE Transactions on Pattern Analysis and Machine Intelligence 14 (2) (1992) 239–256. doi:10.1109/34.121791.

[9] Y. Chen, G. Medioni, Object modelling by registration of multiple range images, Image and Vision Computing 10 (3) (1992) 145–155. doi:10.1016/0262-8856(92)90066-c.

[10] T. Tuna, J. Nubert, Y. Nava, S. Khattak, M. Hutter, X-ICP: Localizability-Aware LiDAR Registration for Robust Localization in Extreme Environments, IEEE Transactions on Robotics 40 (2024) 452–471. doi:10.1109/tro.2023.3335691.

[11] J. Hatleskog, K. Alexis, Probabilistic Degeneracy Detection for Point-to-Plane Error Minimization, IEEE Robotics and Automation Letters 9 (12) (2024) 11234–11241. doi:10.1109/lra.2024.3484153.

[12] F. Pomerleau, F. Colas, R. Siegwart, A Review of Point Cloud Registration Algorithms for Mobile Robotics, Foundations and Trends in Robotics 4 (1) (2015) 1–104. doi:10.1561/2300000035.

[13] C. J. Clopper, E. S. Pearson, The use of confidence or fiducial limits illustrated in the case of the binomial, Biometrika 26 (4) (1934) 404–413. doi:10.1093/biomet/26.4.404.